

## 901 - Online Stock Span

- Span of a stock is # of consecutive days price was  $\leq$  stock on that day

100, 80, 60, 70, 60, 75, 85  
1 1 1 2 1 4 6

$$f(85) = f(75, 4) + f(80, 1) + 1$$

$$\begin{array}{l} f(75, 4) \\ f(80, 1) \\ f(100, 1) \end{array} \rightarrow \begin{array}{l} f(85, 6) \\ f(100, 1) \end{array}$$

|          |          |          |          |
|----------|----------|----------|----------|
|          |          | (60, 1)  |          |
| (60, 1)  | (70, 2)  | (70, 2)  | (75, 4)  |
| (80, 1)  | (80, 1)  | (80, 1)  | (80, 1)  |
| (100, 1) | (100, 1) | (100, 1) | (100, 1) |

Maintain spans with a monotonically decreasing stack to reuse work

price < stack[-1][0] OR stack empty  $\rightarrow$  stack.append((price, 1))

$\rightarrow$  while stack and stack[-1][0]  $\leq$  price:

pop and add to span

append (price, span)